AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST129
Applied calculus

Final Examination
Spring Semester 2025/2026

Date:4 /6 / 2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2:00 Hours**

Instructions:

- Total Marks: 50
- This exam consists of **two** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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Q–1: [5×2 marks] Choose the correct answer:

1.1: which of the following is the simplification of $e^{2\ln x}$

a) $\frac{x}{2}$
b) 2x
c) x^2
d) None of the above

1.2: All solutions of the differential equation $y' = 1.5\ y\ and\ f(0) = 2\ is$:

a) $y=2e^{1.5x}$
b) $y=1.5e^{2x}$
c) $y= 5e^{1.5x}$
d) None of the above

1.3: $\int \frac{e^x-e^{-x}}{2} dx$

a) $\frac{e^x+e^{-x}}{2}+c$
b) $\frac{e^x-e^{-x}}{2}+c$
c) e^x+e^{-x}+c
d)None of the above

1.4: The value of $\int_1^4 \sqrt{x}$ dx is:

a) $\frac{14}{3}$
b) $\frac{16}{3}$
c) $\frac{2}{3}$
d) None of the above

1.5: If $y = x^5 cosx$, then $\frac{dy}{dx} =$

a) $5x^4\sin(x)+x^5\sin(x)$
b) $5x^4\cos(x)-x^5\sin(x)$
c) $-5x^4\sin(x)$
d)None of the above

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Each question is worth 10 marks. Answer the following questions:

Q–2:

a) **[6 marks]** Let $f(x) = x^3 - 3x + 4$
Find the inflection points, if any.

b) **[4marks]** Use implicit differentiation of the equation $-y^2 + y^3 - 4x + x^2 = 0\ to$ find the equation of the tangent line to the graph at the designated point; $P(2,2)$.

Q–3:

a) **[5 marks]**
Find the point on the graph of $y=e^{-x}$ where the tangent line has slope -2.

b) **[5 marks]** use logarithmic differentiation to find the derivative of the function:

$$y= \ln\sqrt{\frac{x^5+1}{x^5+5x+1}}$$

Q–4:

a) **[2×3 marks]** Evaluate the following integrals:

i) $\int_0^2 x^3 - 1\, dx.$

ii) $\int (sec^2(x) - 5\cos(x) + \frac{7}{x})$ dx

b) **[4marks]** Use integration by substitution to evaluate $\int \frac{ln3x}{x}\, dx$

Q–5:

a) [5 marks] Find the area of the region bounded by the graphs of $y = 8x - x^2$ and $y = x^2$

b) [5 marks] Use integration by parts to evaluate $\int x\,Sin(x)\,dx$ dx

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End of questions

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